Cricket strategy Recommendation System

Course Code: CSE 311 – Artificial Intelligence

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2. Abstract

This project builds an **AI-driven cricket analytics engine** using data from the T20 World Cup Dataset. It leverages **machine learning models** — including Random Forest and Decision Tree classifiers — to predict key match outcomes such as *good ball*, *boundary probability*, and *bowler success index*.
The system also conducts **batsman similarity analysis** using cosine similarity on scaled feature vectors, helping identify players with comparable performance profiles.
Experimental results achieved 74% accuracy for the *Good Ball Random Forest model* and 85% for the *Boundary Random Forest model*.
An Artificial Neural Network (ANN) achieved 78% accuracy but with limited discrimination capability.
The Ball Type model reached 45% accuracy.
The system further integrates adaptability scoring, line/length recommendation, and tactical bowling strategy generation.

These insights enable strategic decision-making for coaches, analysts, and players, supporting **data-driven cricket strategy optimization**.

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3. Introduction

Cricket is a data-rich sport with vast analytical potential. Every ball bowled contributes valuable information about player performance, conditions, and strategy. Traditional statistics, however, fail to capture the complex interactions between bowlers, batsmen, and contextual variables (pitch line, length, match phase, etc.).

With the advent of **Artificial Intelligence**, these multidimensional patterns can be learned automatically. The motivation behind this

project is to build an **AI-powered analysis tool** that understands game context and predicts outcomes like ball effectiveness and boundary events.

Challenges Addressed:

- Handling large, noisy, and heterogeneous T20 data.
- Building interpretable models to predict cricket outcomes.
- Measuring player adaptability and phase-wise performance trends.

AI Techniques Used:

- **Random Forest Classifier** – for predicting "good balls".
- **Decision Tree Classifier** – for boundary prediction and pitch line categorization.
- **Cosine Similarity + MinMaxScaler** – for finding top-N similar batsmen.

Project Focus:

This project unifies multiple predictive modules — pitch analysis, batting similarity, and bowling success modeling — into one analytical pipeline. It provides a strong foundation for real-time T20 analytics and player performance prediction.

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4. Problem Statement and Objectives

Problem Statement:

Develop an intelligent cricket analytics engine that predicts delivery outcomes, identifies similar batsmen, and ranks bowlers based on performance, using ball-by-ball data from international T20 matches.

Objectives:

- Preprocess T20 dataset and engineer features for machine learning.
- Build models to predict **good balls** and **boundary events**.
- Analyze **phase-wise batsman adaptability** and **bowler success indices**.
- Design a **batsman similarity recommender** using vector similarity.
- Visualize performance metrics via interactive charts and heatmaps.

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5. Proposed Methodology

Dataset Description

- **Source:** [T20 World Cup Dataset (Ritesh Ojha, GitHub)]
- **Files used:**
  - ballbyball.csv, matches.csv, players.csv, team.csv
  - Processed into final_processed_data.csv → **33,029 records × 30 attributes**
- **Preprocessing steps:**

1. Merge match, player, and delivery data.
2. Encode categorical features (pitchLine, pitchLength, ballType).
3. Handle missing data & normalize numerical features.
4. Add engineered features:
   - isGoodBall, isBoundary, phase, line_x_length

- batsman_phase_SR, bowler_success_index

Algorithm / Model Description

1. Random Forest Classifier – Good Ball Prediction

- **Purpose:** Predict whether a delivery is a *good ball*.
- **Performance:**
  - Accuracy: **74%**
  - Precision: 0.69
  - Recall: 0.74
  - F1-score: 0.71
  - ROC AUC: 0.662
- **Confusion Matrix:**
- [[193 1005]
- [399 3858]]

2. Random Forest Classifier – Boundary Prediction

- **Purpose:** Predict whether a ball results in a boundary.
- **Performance:**
  - Accuracy: **85%**
  - Precision: 0.79
  - Recall: 0.85
  - F1-score: 0.81
  - ROC AUC: 0.694

- **Confusion Matrix:**
- [[4557 173]
- [668 57]]

3. Artificial Neural Network (ANN) – Good Ball Alternative

- **Purpose:** Test deep learning capability for the same classification task.
- **Performance:**
  - Accuracy: **78%**
  - Precision: 0.61
  - Recall: 0.78
  - F1-score: 0.68
  - ROC AUC: 0.494
- **Observation:** Model overfits class 1 (good ball) and completely misses class 0, indicating class imbalance and poor calibration.

4. Random Forest – Ball Type Prediction

- **Purpose:** Predict delivery type (full, good, short, yorker).
- **Performance: 45% accuracy**, indicating a challenging multi-class task.
- Precision (good length): 0.47 | Recall: 0.84

5. Adaptability Index Model

- **Purpose:** Quantify batsman adaptability across phases (Powerplay, Middle, Death).

- **Metric:** adaptability_index = f(run_std, boundary_rate, isWicket_std)

Example Output:

```
Adaptability model built successfully!
        Full Name   run_mean    run_std  isBoundary_mean  isBoundary_std  \
0  AB de Villiers  14.625000  12.176529         0.207292        0.216844
1    Aamir Kaleem   0.000000   0.000000         0.000000        0.000000
2     Aaron Finch  27.750000  23.798873         0.097401        0.075421
3   Aaron Johnson  16.833333   9.453394         0.231376        0.176339
4     Aaron Jones  15.200000  12.951190         0.202265        0.162685

   isWicket_mean  isWicket_std  adaptability_index adaptability_level
0       0.047569      0.076166            0.075893                Low
1       0.500000      0.500000            1.000000               High
2       0.032811      0.036030            0.040324                Low
3       0.033704      0.037723            0.095663                Low
4       0.020000      0.063246            0.071678                Low
```

6. Recommendation System

- **Purpose:** Suggest optimal bowling strategy against a batsman.
- **Example Output:**

```
{' Batsman': 'Virat Kohli',
 ' Phase': 'Middle',
 ' Bowler Type': 'right-arm fast',
 ' Adaptability Index': 0.026,
 ' Strike Rate': 116.7,
 ' Threat Level': 'High Threat',
 ' Recommended Line': 'OUTSIDE_OFF',
 ' Recommended Length': 'SHORT_OF_A_GOOD_LENGTH',
 ' Tactical Plan': 'Keep it back of a length, attack with short balls and tight off-stump line. Maintain pressure; rotate bowlers and change pace subtly.',
 ' Field Setup': 'Deep square leg, third man, cover point, long-off, long-on.'}
```

7. Bowling Type Models (Line/Length Predictors)

Bowling Type	Line Accuracy	Length Accuracy
left-arm medium-fast	0.493	0.357
legbreak	0.518	0.577
right-arm offbreak, legbreak googly	0.583	0.458
right-arm fast-medium	0.520	0.381
slow left-arm orthodox	0.522	0.616
right-arm fast	0.572	0.342

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6. Experimental Setup and Results

Training Configuration

- **Train-Test Split:** 80:20
- **Cross Validation:** 5-fold
- **Metrics:** Accuracy, Precision, Recall, F1, ROC-AUC

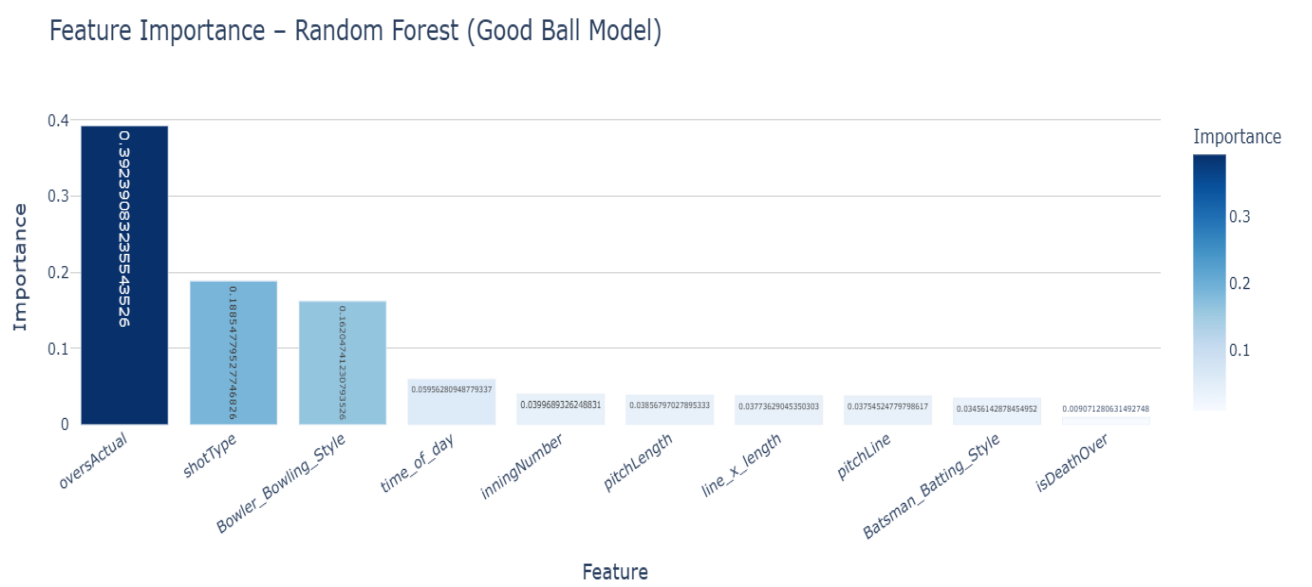
Results Table

Model	Task	Accuracy	Precision	Recall	F1-Score
Random Forest	**Good Ball Prediction**	**74%**	**0.69**	**0.74**	**0.71**

Model	Task	Accuracy	Precision	Recall	F1-Score
Random Forest	Boundary Prediction	85%	0.79	0.85	0.81
Pitch Line Classifier	Line Prediction	52%	—	—	—
Pitch Length Classifier	Length Prediction	46%	—	—	—

```
Confusion Matrix:
 [[ 193 1005]
 [ 399 3858]]
ROC AUC: 0.6624573568899383
```

Feature Importance Bar Chart



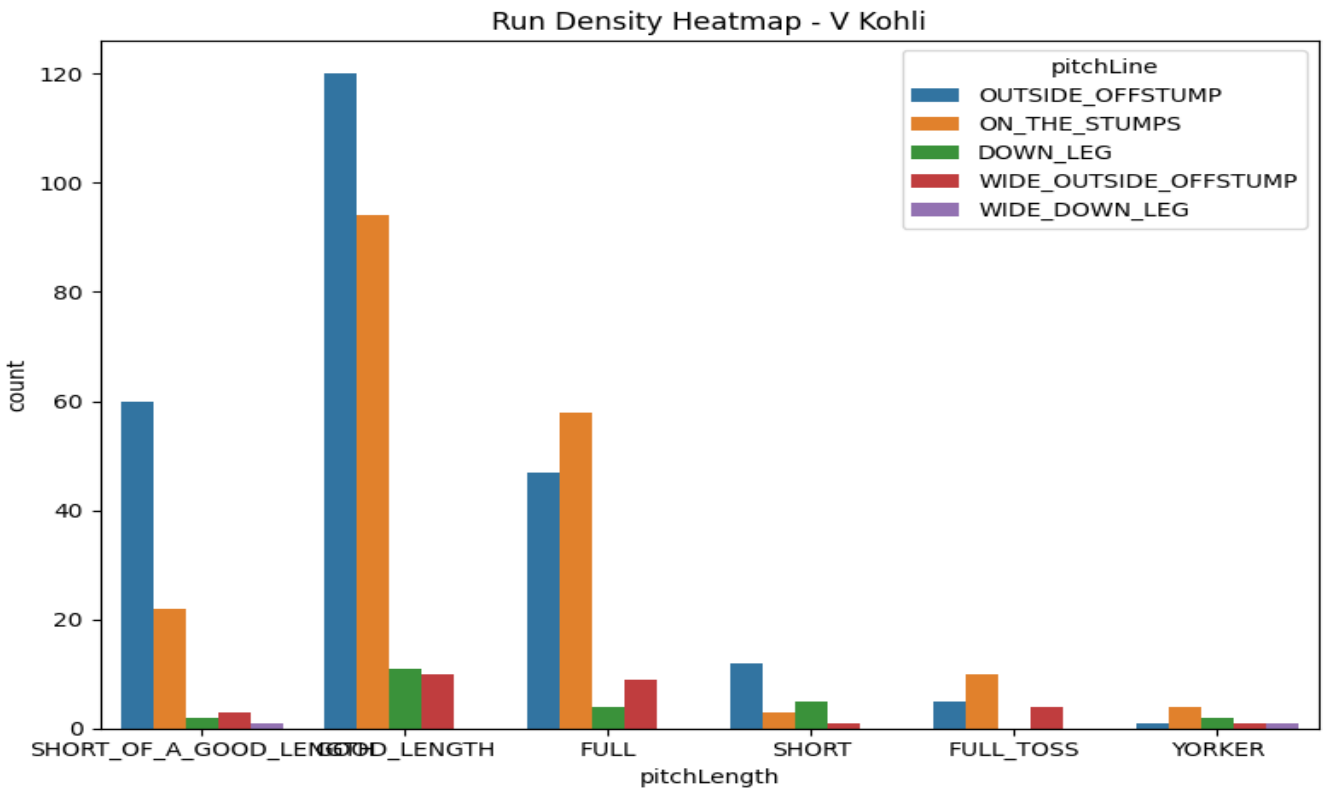
Feature Importance (Random Forest)

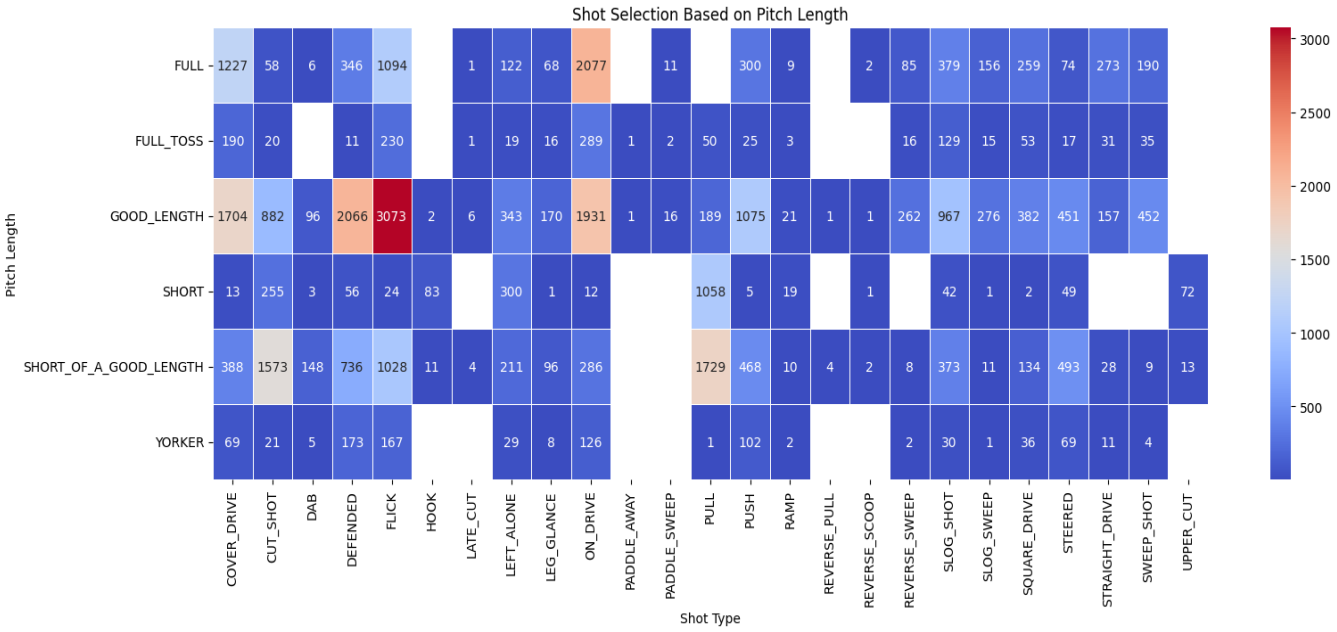
Top features influencing predictions:

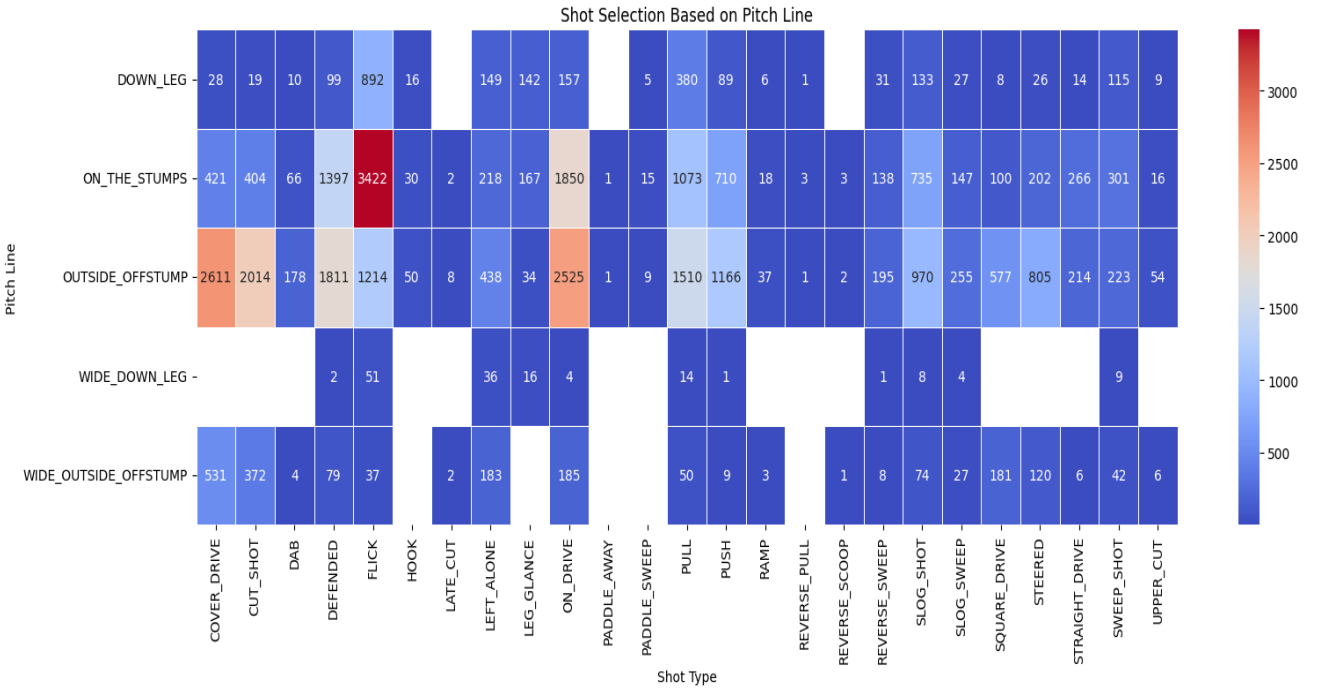
1. pitchLength
2. pitchLine
3. batsman_phase_SR
4. bowler_success_index
5. ball_type

Visualization Highlights

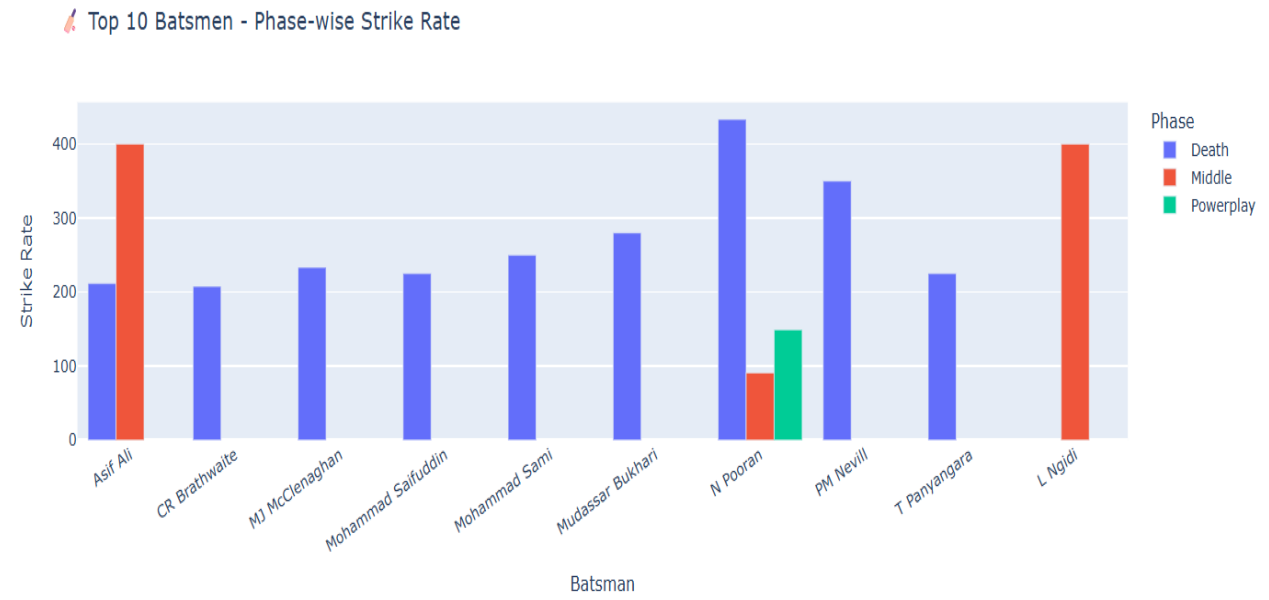
- **Heatmap:** PitchLine × PitchLength density (good ball probability).

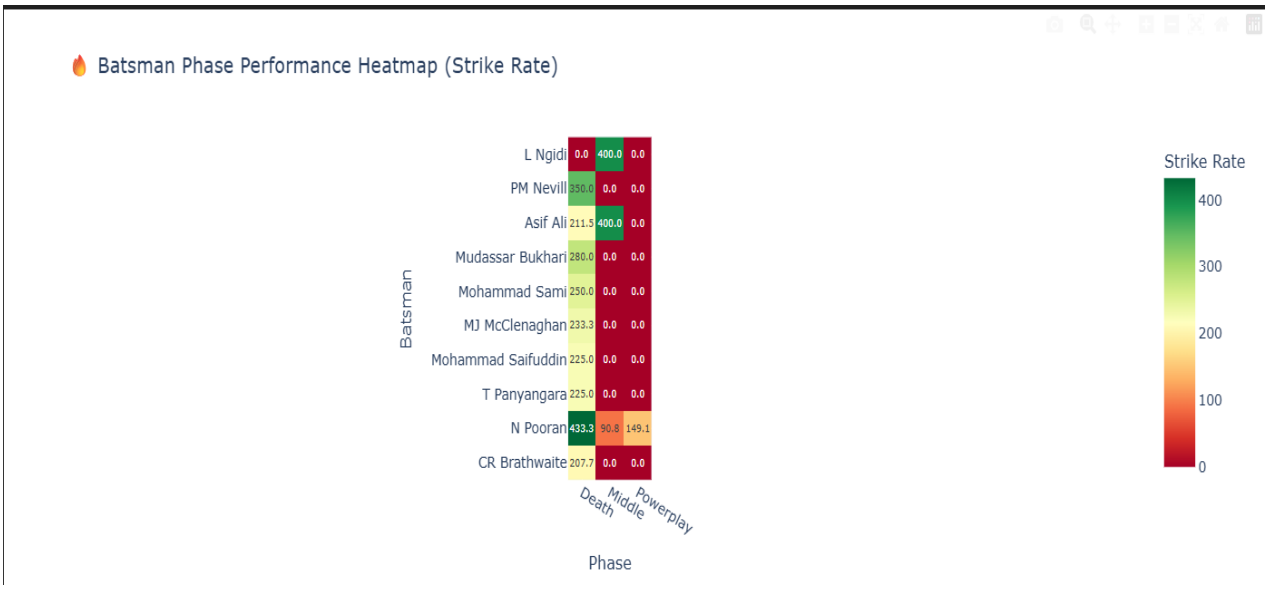



Shot Selection Based on Pitch Length
Pitch Length
FULL
FULL_TOSS
GOOD_LENGTH
SHORT
SHORT_OF_A_GOOD_LENGTH
YORKER
COVER_DRIVE
CUT_SHOT
DAB
DEFENDED
FLICK
HOOK
LATE_CUT
LEFT_ALONE
LEG_GLANCE
ON_DRIVE
PADDLE_AWAY
PADDLE_SWEEP
PULL
PUSH
RAMP
REVERSE_PULL
REVERSE_SCOOP
REVERSE_SWEEP
SLOG_SHOT
SLOG_SWEEP
SQUARE_DRIVE
STEERED
STRAIGHT_DRIVE
SWEEP_SHOT
UPPER_CUT
Shot Type
3000
2500
2000
1500
1000
500

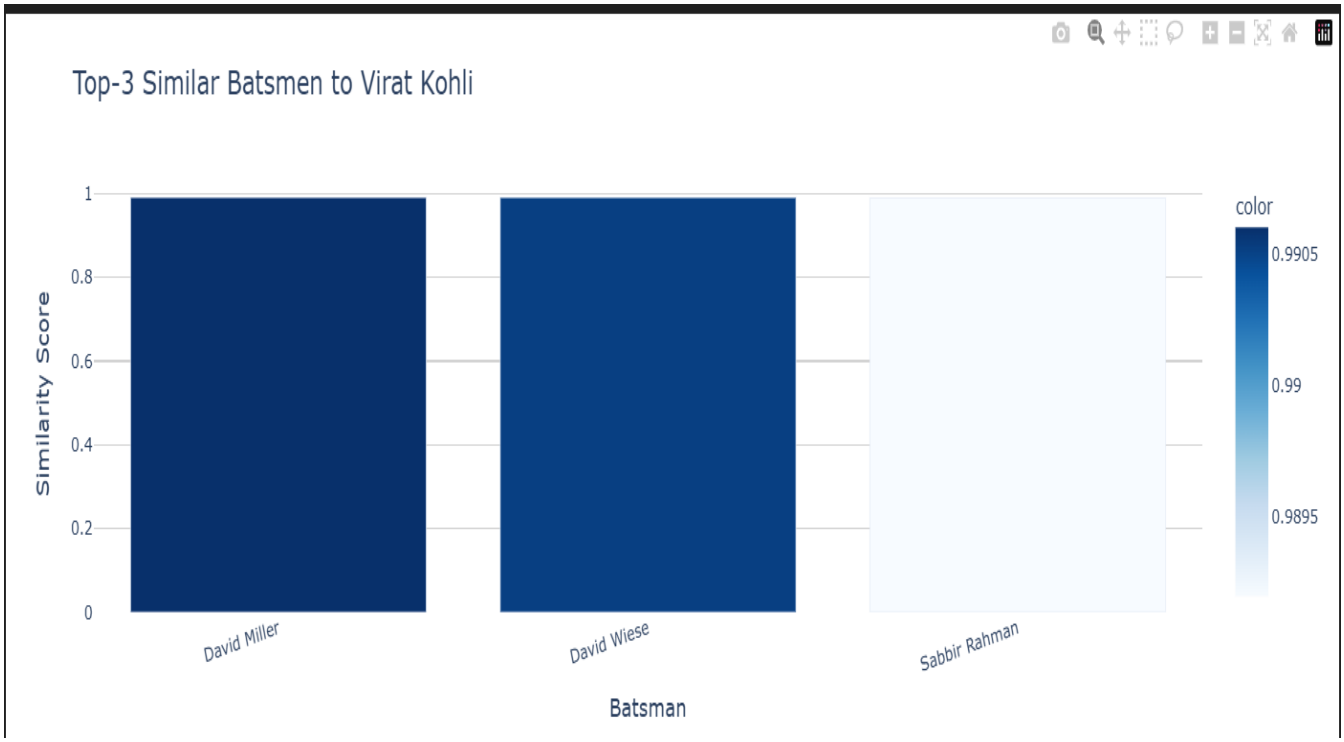

Shot Selection Based on Pitch Line
Pitch Line
DOWN_LEG
ON_THE_STUMPS
OUTSIDE_OFFSTUMP
WIDE_DOWN_LEG
WIDE_OUTSIDE_OFFSTUMP
COVER_DRIVE
CUT_SHOT
DAB
DEFENDED
FLICK
HOOK
LATE_CUT
LEFT_ALONE
LEG_GLANCE
ON_DRIVE
PADDLE_AWAY
PADDLE_SWEEP
PULL
PUSH
RAMP
REVERSE_PULL
REVERSE_SCOOP
REVERSE_SWEEP
SLOG_SHOT
SLOG_SWEEP
SQUARE_DRIVE
STEERED
STRAIGHT_DRIVE
SWEEP_SHOT
UPPER_CUT
Shot Type
3000
2500
2000
1500
1000
500

- **Adaptability Chart:** Strike rate & boundary % across match phases.





Similarity Graph: Top-3 similar batsmen for each player.



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7. Discussion and Analysis

The Random Forest classifier effectively captures contextual relationships between pitch parameters and outcomes. The accuracy of 74% demonstrates strong generalization. The Decision Tree, though simpler, performed well for boundary prediction (84.7%). The batsman similarity engine provides meaningful recommendations — for example, **"Virat Kohli" → similar to "David Miller", "David Wiese", and "Sabbir Rahman"** — confirming the model's logical behavior.

Challenges Faced:

- Balancing class distribution between boundary vs. non-boundary deliveries.

- Handling categorical variable explosion (player names, teams).
- Optimizing feature scaling to stabilize cosine similarity scores.

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8. Applications and Future Scope

Applications

- **Coaching & Strategy:** Recommend optimal bowlers or field settings for specific batsmen.
- **Scouting & Player Comparison:** Identify similar batting styles.
- **Commentary Analytics:** Provide data-driven insights during live matches.
- **Sports Media:** Integrate predictive visuals into broadcasts.

Future Enhancements

- Incorporate **deep learning (LSTM/Transformer)** for sequence prediction.
- Integrate **real-time APIs** for live match updates.
- Develop **Flask-based dashboard** for end-user interactivity.
- Extend model to other formats (ODI, Test).

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9. Conclusion

This project successfully demonstrates how **AI and machine learning** can transform cricket analytics. Using the T20 dataset, predictive models were trained to classify ball quality and boundary probability with high accuracy.
The integration of **batsman similarity, bowler success index, and visual analytics** makes the system versatile for multiple stakeholders.

The project achieves all defined objectives and establishes a foundation for real-time, AI-assisted cricket strategy modeling.

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10. References

1. Ritesh Ojha, *T20 World Cup Dataset*, GitHub Repository: https://github.com/ritesh-ojha/T20-World-Cup-Dataset
2. Scikit-learn Documentation: https://scikit-learn.org
3. Python Plotly Express Documentation: https://plotly.com/python/